

# Electronics for Embedded Systems notes

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## 1 Lesson 23-09-26

### 1.1 Introduction

The course is meant to give an introduction to *embedded systems*, after a review of analog and digital electronics. The topics covered are:

- Memories;
- (Programmable Logic Devices) (PLDs);
- Interconnection protocols and I/O;
- Embedded processors and peripherals;
- **DA** (*Digital-to-Analog*) and **AD** (*Analog-to-Digital*) conversion;
- Power management.

### 1.2 Review

Let's begin the review of analog and digital electronics. The review will cover:

- Electrical components, including:
  - **Passive components**, as in *resistors*, *capacitors* and *inductors*;
  - **Active components**, as in *diodes* and *transistors*;
  - **Operational amplifiers**.
- Digital electronics, including *logic gates*, *digital functions*, **FSMs** (*Finite State Machines*) and the *VHDL* language.

#### 1.2.1 Passive components

**Resistors** offer *resistance* to electric current, measured in *Ohms* ( $\Omega$ ). By Ohm's law:

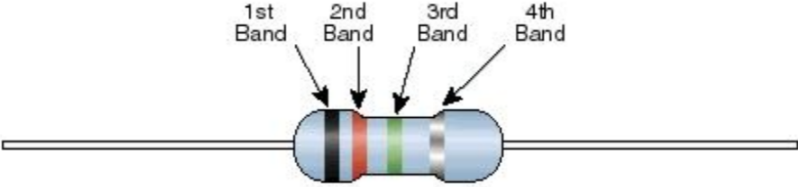
$$v_R(t) = Ri_R(t), \quad V_R(s) = RI_R(s)$$

with  $s = \sigma + j\omega$  in the Laplace domain. A definition via *conductance*, measured in *Siemens* ( $S$ ), is also possible:

$$i_R(t) = Gv_R(t), \quad g = \frac{1}{R}$$

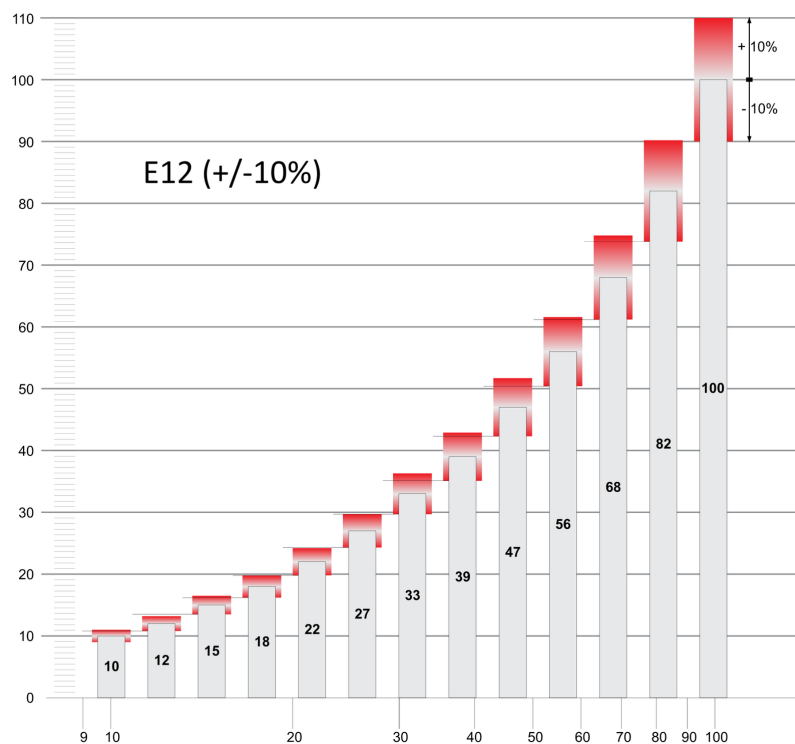
Resistor response doesn't depend on the frequency of periodic signals (all signals get attenuated) the same. This applies at least partly to real resistors, and we will assume the ideal model from now onwards.

It can also be useful to remember the code used to distinguish different resistance values.

**Standard EIA Color Code Table 4 Band:  $\pm 2\%$ ,  $\pm 5\%$ , and  $\pm 10\%$** 


| Color  | 1st Band<br>(1st figure) | 2nd Band<br>(2nd figure) | 3rd Band<br>(multiplier) | 4th Band<br>(tolerance) |
|--------|--------------------------|--------------------------|--------------------------|-------------------------|
| Black  | 0                        | 0                        | $10^0$                   |                         |
| Brown  | 1                        | 1                        | $10^1$                   |                         |
| Red    | 2                        | 2                        | $10^2$                   | $\pm 2\%$               |
| Orange | 3                        | 3                        | $10^3$                   |                         |
| Yellow | 4                        | 4                        | $10^4$                   |                         |
| Green  | 5                        | 5                        | $10^5$                   |                         |
| Blue   | 6                        | 6                        | $10^6$                   |                         |
| Violet | 7                        | 7                        | $10^7$                   |                         |
| Gray   | 8                        | 8                        | $10^8$                   |                         |
| White  | 9                        | 9                        | $10^9$                   |                         |
| Gold   |                          |                          | $10^{-1}$                | $\pm 5\%$               |
| Silver |                          |                          | $10^{-2}$                | $\pm 10\%$              |

Resistor tolerances are to be taken into account, as they might drive resistances in our circuits out of the feasible margins. Because of this, we prefer using standard values in *E series*. E series is a system of preferred numbers (also called preferred values) derived for use in electronic components. It consists of the E3, E6, E12, E24, E48, E96, and E192 series, where the number after the "E" designates the quantity of logarithmic value "steps" per decade.



**Capacitors** store energy as voltage potential between conductive armatures separated by a dielectric. *Capacity*, measured in *Farad* ( $F$ ) is defined as:

$$C = \frac{Q}{V}$$

The laws that govern capacitors in the time domain are differential equations:

$$i_C(t) = C \frac{d}{dt} v_C(t), \quad v_C(t) = \frac{1}{C} \int_0^t i_C(\tau) d\tau$$

It is then easier to move to the Laplace domain, where we can simply express integration as division by  $s$ :

$$V_C(s) = \frac{I_C(s)}{sC}$$

In regards to periodic signals, the behaviour of capacitors depends on frequency:

- When  $f \rightarrow 0$  it behaves as an *open circuit*;
- When  $f \rightarrow \infty$  it behaves as a *short circuit*;

Many types of capacitors are available:

- *Electrolytic* capacitors, which are polar;
- *Ceramic* capacitors, which are not polar.

To read the value of a capacitor we have quite a complex system:

- Two digits: it is the value in  $pF$ ;
- Three digits: the first two are the value in  $pF$ , and the third is the multiplier; If the third digit is 8 the multiplier is  $10^{-2}$ , if it is 9 the multiplier is  $10^{-1}$ ;
- A digit, a character and a digit: the two digits are the one before and after the decimal point, the character is the unit of measure;
- A character follows the last digit indicates the tolerance, with  $J = \pm 5\%$ ,  $K = \pm 10\%$ ,  $M = \pm 20\%$ .

**Inductors** store energy as magnetic field in a winding of conductive wire around a metal core. *Inductance*, measured in *Henry* ( $H$ ) is defined as:

$$L = \frac{\Phi}{I}$$

The laws that govern inductors in the time domain are differential equations:

$$v_L(t) = L \frac{d}{dt} i_L(t), \quad i_L(t) = \frac{1}{L} \int_0^t v_L(\tau) d\tau$$

It is then easier to move to the Laplace domain, where we can simply express derivation as multiplication by  $s$ :

$$V_L(s) = sLI_L(s)$$

In regards to periodic signals, the behaviour of inductors also depends on frequency, on the contrary of capacitors:

- When  $f \rightarrow \infty$  it behaves as an *open circuit*;
- When  $f \rightarrow 0$  it behaves as a *short circuit*;

### 1.2.2 Voltage and current dividers

We show a chart showcasing a number of elementary divider circuits which we can use.

- Voltage divider with two resistances

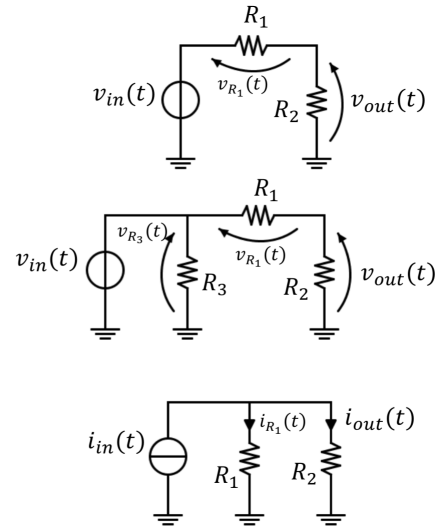
- $v_{out}(t) = \frac{R_2}{R_1 + R_2} v_{in}(t) = \frac{g_1}{g_1 + g_2} v_{in}(t)$
  - $v_{R_1}(t) = \frac{R_1}{R_1 + R_2} v_{in}(t) = \frac{g_2}{g_1 + g_2} v_{in}(t)$

- Voltage divider with three resistances

- $v_{out}(t) = \frac{R_2}{R_1 + R_2} v_{in}(t) = \frac{g_1}{g_1 + g_2} v_{in}(t)$
  - $v_{R_3}(t) = v_{in}(t)$

- Current divider

- $i_{out}(t) = \frac{R_1}{R_1 + R_2} i_{in}(t) = \frac{g_2}{g_1 + g_2} i_{in}(t)$
  - $i_{R_1}(t) = \frac{R_2}{R_1 + R_2} i_{in}(t) = \frac{g_1}{g_1 + g_2} i_{in}(t)$



Let us note in the second example (voltage divider with three resistances) that nothing changes by adding the third resistor.  $V_{out}(t)$  is exactly the same, as the potential seen by  $R_1$  on its left side doesn't change (and the potential across  $R_3$  is the same as  $v_{in}(t)$ ).

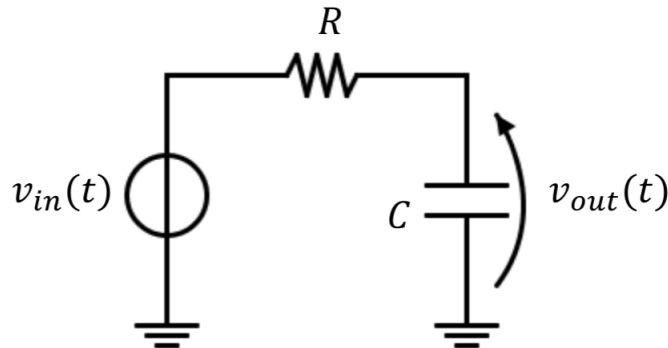
What actually does change in the example is the resistance: we have effectively built a current divider, where the input current is:

$$i_{in}(t) = \frac{v_{in}(t)}{R_1 + R_2} + \frac{v_{in}(t)}{R_3}$$

Lastly, let's note that all the equations in the Laplace domain are the same, changing the variables accordingly.

### 1.2.3 Simple low pass filters

A simple low pass filter can be seen as a special case of a divider:



By writing the *Kirchoff Voltage Law* (KVL) of the circuit, we get:

$$v_{in}(t) - R \cdot C \frac{d}{dt} v_{out}(t) - v_{out}(t) = 0$$

with the first term being the voltage gain on the power supply, the second term being the voltage drop on the resistor, and the last term being the output voltage taken across the capacitor.

**Time domain solution** This is a first order linear differential equation, which we can express as:

$$\frac{d}{dt}v_{out}(t) + \frac{1}{\tau}v_{out}(t) = \frac{1}{\tau}v_{in}(t), \quad \tau = RC$$

We refer to  $\tau = RC$  as the *time constant* of the circuit. A general solution is:

$$v_{out}(t) = e^{-\frac{t}{\tau}} \left( k + \int \left( \frac{v_{in}(t)}{\tau} e^{\frac{t}{\tau}} \right) dt \right)$$

It can be useful to carry out the full calculation. We start from the general differential equation obtained from KVL:

$$v_{in}(t) - RC \frac{d}{dt}v_{out}(t) - v_{out}(t) = 0$$

We can easily rearrange this into a standard first-order form:

$$\frac{d}{dt}v_{out}(t) + \frac{1}{RC}v_{out}(t) = \frac{1}{RC}v_{in}(t)$$

Now we take a function  $\mu(t)$  constructed such that:

$$\mu(t) \frac{1}{RC} = \frac{d}{dt}\mu(t)$$

While the purpose of this might at this point be unclear, we can clearly see that a function that satisfies the equation is:

$$\mu(t) = e^{\frac{t}{RC}}$$

Let's multiply both sides of the original differential equation by  $\mu(t)$ :

$$\mu(t) \frac{d}{dt}v_{out}(t) + \mu(t) \frac{1}{RC}v_{out}(t) = \mu(t) \frac{1}{RC}v_{in}(t)$$

By applying the property above this leaves us with:

$$\mu(t) \frac{d}{dt}v_{out}(t) + \frac{d}{dt}\mu(t)v_{out}(t) = \frac{\mu(t)}{RC}v_{in}(t)$$

It's clear that the left hand side is the expansion of the derivative of a product, so we can write:

$$\frac{d}{dt}(\mu(t)v_{out}(t)) = \frac{\mu(t)}{RC}v_{in}(t)$$

Recall that we were interested in  $v_{out}(t)$ . We can take it out the derivative by integrating both sides:

$$\mu(t)v_{out}(t) = \int \frac{\mu(t)}{RC}v_{in}(t) + k$$

where the antiderivative operation requires that we include a  $\pm k$  factor (sign indifferent).

At this point we just have to make  $v_{out}(t)$  explicit and substitute our original definition of  $\mu(t)$ :

$$v_{out}(t) = \frac{1}{\mu(t)} \left( \int \frac{\mu(t)}{RC} v_{in}(t) + k \right) = e^{-\frac{t}{RC}} \left( k + \int \left( \frac{v_{in}(t)}{RC} e^{\frac{t}{RC}} \right) dt \right)$$

Which is our original expression (given  $\tau = RC$ ).  $\square$

With  $v_{in}(t)$  taken as a step of amplitude  $V_M$ , or equivalently by taking the solution given with zero initial conditions and a constant power source, we get the usual form:

$$v_{out}(t) = V_M(1 - e^{-\frac{t}{\tau}})$$

where the time constant determines the time taken to reach saturation. We recall that in one time constant we reach around 63.2% of full saturation, and in three time constants we reach 95%.

**Laplace domain solution** Let's carry out the same calculation in the Laplace domain. By expressing the voltage divider we get:

$$V_{out}(S) = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} V_{in}(S) = \frac{1}{sRC + 1} V_{in}(s)$$

where in the first expression we simply applied the voltage divider law to the impedance definitions of the capacitor and the resistor. We can then derive a *transfer function*:

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{sRC + 1}$$

We can shift to the Fourier domain, or in equivalent terms take a projection of the Laplace domain in only the frequency direction, by setting  $s = j\omega$ :

$$\frac{V_{out}(j\omega)}{V_{in}(j\omega)} = \frac{1}{j\omega RC + 1}, \quad \left| \frac{V_{out}(j\omega)}{V_{in}(j\omega)} \right| = \frac{1}{\sqrt{\omega^2 R^2 C^2 + 1}}$$

A *pole* in this expression is a zero of the denominator, so:

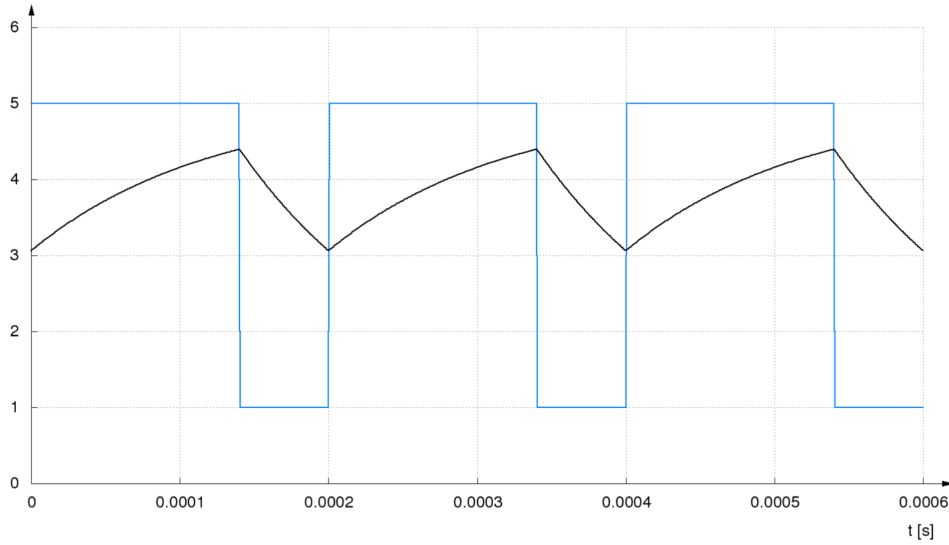
$$s_p = -\frac{1}{RC}, \quad \omega_p = \frac{1}{RC}$$

with the frequency representation being bilateral, as always. Let's note that the *pole pulsation*  $\omega_p$  is not really a zero of the denominator, but just the characteristic frequency of the system, which we find useful for plotting and solving circuit responses.

**Time domain response** Fixing  $V_{in}$  to a periodic square wave, let's compute the time domain response. Each fragment in the response is an exponential:

$$v_{out}(t) = A + B e^{-\frac{t}{\tau}}$$

where  $\tau = RC$ .



We can proceed with the following approach:

1. Evaluate the waveforms at the extreme, fixing initial conditions;
2. Determine A and B for the rising and falling cases;
3. Determine the maximum and minimum voltages.

And the result we obtain should be valid at the steady-state. We say *steady-state*, as when the oscillators first begins oscillating there will be a transient, which will temporarily not obey to our prediction.

Let's start with the rising edge. By remembering our first equation, we can compute  $V_{out}$  at the first time instant:

$$V_{out}^r(0) = A + B = V_{min}$$

If at this point the square wave never went to a low value, the capacitor would continue charging and we would approach saturaiton (at  $V_H$ , the maximum value of the square wave). From this we can derive  $A = V_H$ , as the  $B$  term vanishes to 0 approaching infinity, and it follows that  $B = V_{min} - V_H$ . This gives us:

$$V_{out}^r(t) = V_H + (V_{min} - V_H)e^{-\frac{t}{\tau}}$$

What happens in reality is that at some time  $T_H$  we switch to the falling edge, reaching a maximum value  $V_{max}$ .

$$V_{out}^r(T_H) = A + Be^{-\frac{T_H}{\tau}} = V_H + (V_{min} - V_H)e^{-\frac{T_H}{\tau}} = V_{max}$$

To fix  $V_{min}$  and  $V_{max}$ , we switch to analyzing the falling edge. With similar calculations, we can derive the equation:

$$V_{out}^f(T_L) = V_L + (V_{max} - V_L)e^{-\frac{T_L}{\tau}} = V_{min}$$

with  $V_L$  at the lowest point of the square wave.

We can easily solve the resulting system of equations, to figure out the minimum and maximum bounds of the response of the circuit:

$$\begin{cases} V_H + (V_{min} - V_H)e^{-\frac{T_H}{\tau}} = V_{max} \\ V_L + (V_{max} - V_L)e^{-\frac{T_L}{\tau}} = V_{min} \end{cases}$$

By setting:

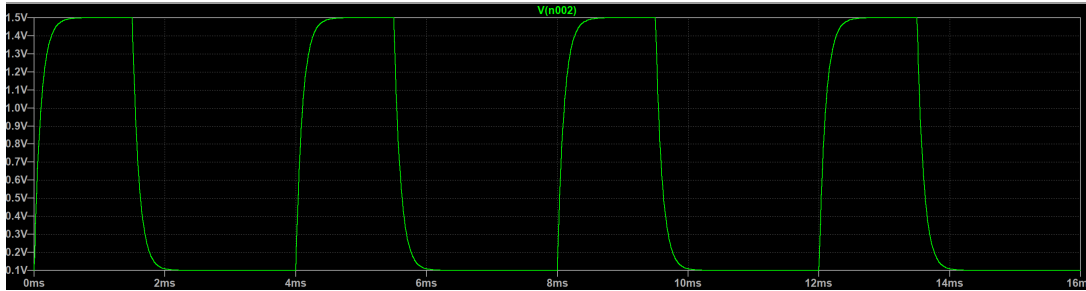
$$\alpha = e^{-\frac{T_H}{T}}, \quad \beta = e^{-\frac{T_L}{T}}$$

we get the solutions (calculations omitted because they're boring):

$$V_{max} = \frac{V_H (1 - \alpha) + \alpha V_L (1 - \beta)}{1 - \alpha\beta}, \quad V_{min} = \frac{\beta V_H (1 - \alpha) + V_L (1 - \beta)}{1 - \alpha\beta}$$

We can then trace the actual response by drawing exponential branches with the time constant given by  $\tau = RC$ .

Lastly, let's see an LTSPICE (simulation program) plot of the square wave response of this circuit, which confirms our mathematical model:



**Frequency domain response** In the frequency domain, we can consider the effect on a periodic signal such as a sine wave. Let's have:

$$V_{in}(t) = M \sin(\omega t) = M \sin(2\pi f t)$$

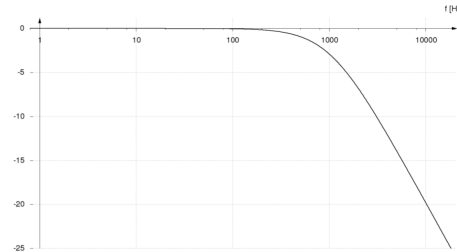
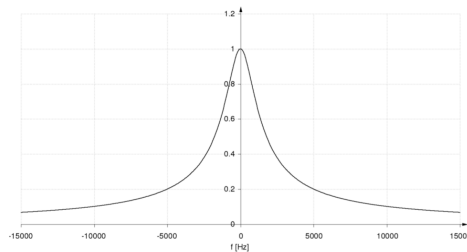
In a linear system at the steady state (which we have) if the input is sinusoidal, the output is sinusoidal at the same frequency: only the amplitude and the phase can be shifted:

$$V_{out}(t) = M' \sin(\omega t + \phi)$$

By using the Fourier domain equations calculated earlier, we can actually express  $M'$  and  $\phi$ :

$$M' = M \frac{\omega_p}{\sqrt{\omega^2 + \omega_p^2}}, \quad \phi = -\tan^{-1} \left( \frac{\omega}{\omega_p} \right)$$

For arbitrary frequencies, we can plot the linear and logarithmic Bode diagrams of amplitude:



Let us note that at  $f_p = \frac{\omega_p}{2\pi}$  we have the  $-3$  dB band for our transfer function, which we can also refer to as the *cutoff frequency* or *pole frequency* of the lowpass.

### 1.2.4 Superposition of effects

#### Theorem 1.1: Superposition of effects

In a linear system with more than one input (independent ideal sources, voltage or current), the output is the algebraic sum of the effects of each input with the other set to 0.

This means:

- Voltage source at zero  $\rightarrow$  short circuit;
- Current source at zero  $\rightarrow$  open circuit;
- Dependent sources must not be set to 0.

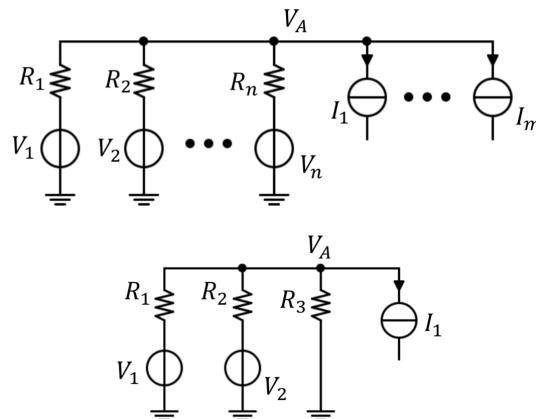
Superposition applies to *linear* systems, that is circuits which feature:

- Sources (independent and dependent), resistors, capacitors, inductors, transformers;
- Non linear system

If diodes, transistors, operational amplifiers or other active elements are present, the superposition of effects is not applicable. In these cases we need to first *linearize* the system around a bias point.

### 1.2.5 Millman's theorem

Let us take the circuit:



#### Theorem 1.2: Millman's Theorem

In the general network on the right, the voltage  $V_A$  is given by the expression:

$$V_A = \frac{\sum_{i=1}^n \frac{V_i}{R_i} - \sum_{j=1}^m I_j}{\sum_{i=1}^n \frac{1}{R_i}}$$

The theorem can be proved using the superposition of effects and voltage dividers with conductances. Let us work through it.

The idea is to:

1. Calculate the contribution on voltage  $V_A$  given by just a voltage generator  $V_j$ ; then we can sum the contribution over all voltage generators to get the cumulative voltage generator contribution, which we'll call  $V_A^a$ ;

2. Calculate the contribution on voltage  $V_A$  given by just a current generator  $I_j$ ; then we can sum the contribution over all current generators to get the cumulative current generator contribution, which we'll call  $V_A^b$ .

Superposition is applied in both steps, and one final time to sum the contributions of the voltage and current generators and get the final expression for  $V_A$ .

- For the voltage generators, we take the contribution of a single voltage generator  $V_j$  as:

$$V_A^{a_j} = \frac{\parallel_{i \neq j}^n R_i}{\parallel_{i \neq j}^n R_i + R_j} V_j$$

where  $\parallel_{i \neq j}^n R_i$  is notation soup for the parallel of all resistance but  $R_j$ .

We can sum over these to get the total voltage generator contribution  $V_A^a$ :

$$V_A^a = \sum_{j=1}^n \frac{\parallel_{i \neq j}^n R_i}{\parallel_{i \neq j}^n R_i + R_j} V_j$$

This sum is a bit hard to compute, as the denominator changes at each value of  $j$  and we don't have a clean way to expand it. Things would be quite a bit easier if we had the same denominator at each value of  $j$ . We can make this happen by dividing the numerator and the denominator by  $R_j \parallel_{i \neq j}^n R_i$ , which gives:

$$V_A^a = \sum_{j=1}^n \frac{\frac{1}{R_j}}{\frac{1}{R_j} + \frac{1}{\parallel_{i \neq j}^n R_i}} V_j$$

To our convenience, we note that  $\frac{1}{\parallel_{i \neq j}^n R_i}$  is equal to  $\sum_{i \neq j}^n \frac{1}{R_i}$ , so:

$$V_A^a = \sum_{j=1}^n \frac{\frac{1}{R_j}}{\frac{1}{R_j} + \sum_{i \neq j}^n \frac{1}{R_i}} V_j = \sum_{j=1}^n \frac{\frac{1}{R_j}}{\sum_{i=1}^n \frac{1}{R_i}} V_j = \frac{\sum_{i=1}^n \frac{V_i}{R_i}}{\sum_{i=1}^n \frac{1}{R_i}}$$

- Current generators are thankfully simpler to handle. We take the contribution of a single current generator  $I_j$  as:

$$V_A^{b_j} = I_j \parallel_{i=1}^n$$

We can sum over these to get the total voltage generator contribution  $V_A^b$ :

$$V_A^b = \sum_{j=1}^m I_j \parallel_{i=1}^n = \sum_{j=1}^m \frac{I_j}{\sum_{i=1}^n \frac{1}{R_i}} = \frac{\sum_{j=1}^m I_j}{\sum_{i=1}^n \frac{1}{R_i}}$$

At this point we just sum to get the final result. □

### 1.2.6 Active components

**Diodes** are non-linear active components, built today using solid state technologies, and more specifically *PN junctions* (as opposite to tubes, which were historically used).

By combining two doped regions of semiconductors (usually Silicon), respectively P (boron) and N (arsenic or phosphorous) doped, we obtain an initial condition where:

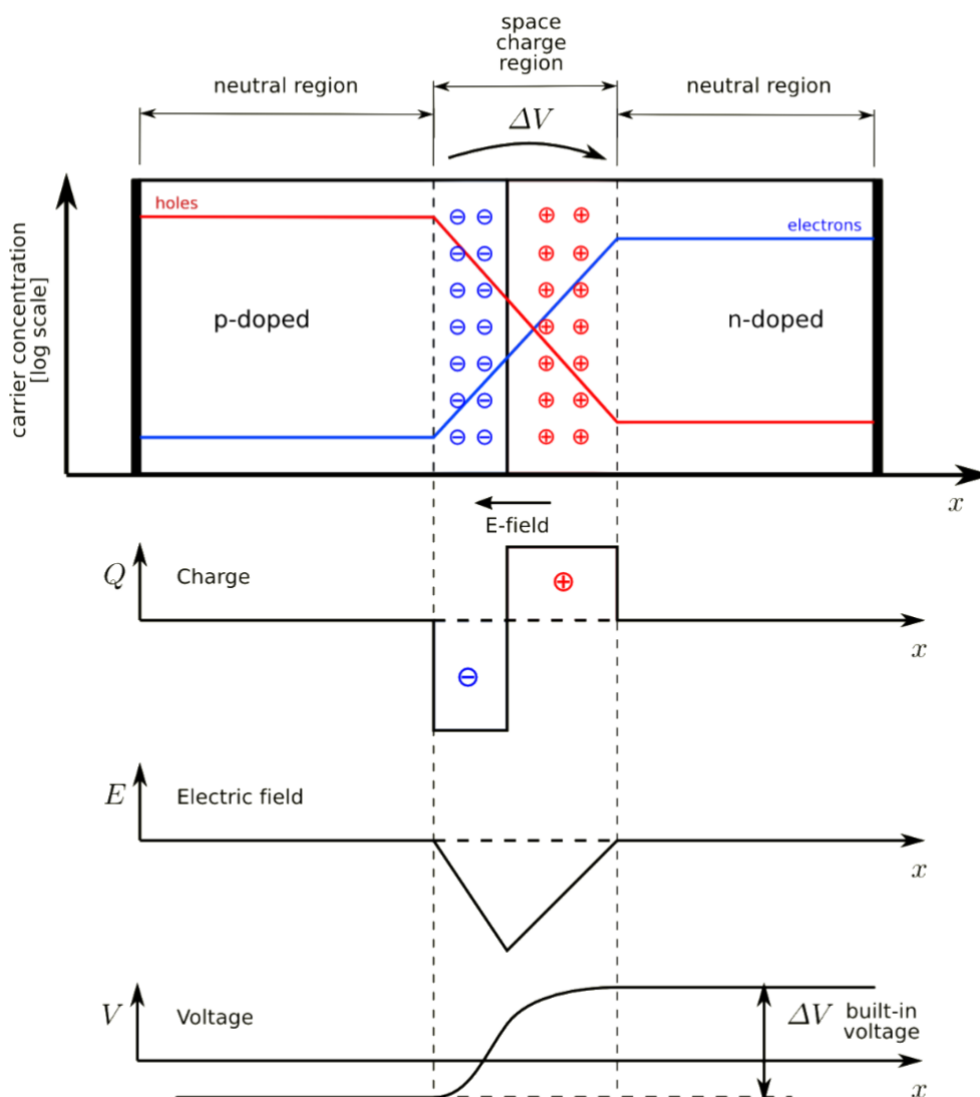
- Electrons from the n side flow by diffusion to the p side: fixed positive ions are left in the n side.

- Holes from the p side flow by diffusion to the n side: fixed negative ions are left in the p side.

At thermal equilibrium:

- Fixed ions in the two sides create an electrical field which opposes the further diffusion of electrons and holes, this is called a *depletion layer* or *space charge region*;
- A built-in voltage  $V_{bi}$  is established;
- The built-in voltage opposes current, acting as a barrier, or *threshold*, which must be overcome in order to have current flowing.

The usual value of the voltage needed to get current flowing through a diode is around  $V_\gamma \approx 0.7V$ .



Applying a voltage to the junction, that is applying a *forward bias*, we get multiple effects around  $V_\gamma$ :

- No current flows if  $v_D < V_\gamma$ ;
- Current starts flowing if  $v_D > V_\gamma$ .

In *reverse bias*, no current flows and depletion layer width increases. For large reverse voltages, we see *breakdown* of the junction, followed by a large reverse current.

This effect is the combined result of *tunneling effect* or *avalanche effect*.

There is no damages to the diode, but the large current may increase power dissipation and heat, causing damages. This effect is exploited in Zener diodes, which break at specific reference voltages.

## 2 Lesson 24-09-26

We continue treating *active components*, starting from the **current-voltage** characteristic of the Diode:

### Definition 2.1: Schottky's law

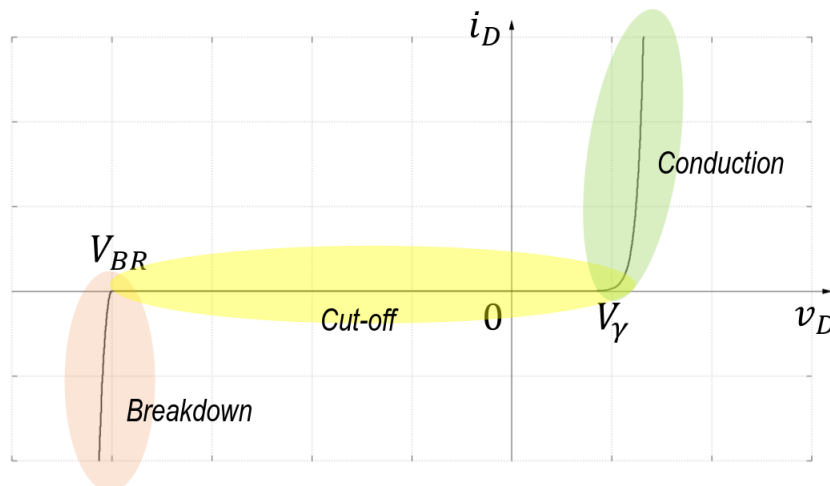
The current-voltage characteristic of a diode is:

$$i_D(t) = I_S \left( e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$$

Where:

- $I_S$  is the *saturation current*, interpreted as the maximum reverse current we can force through the diode;
- $\eta$  is the *ideality factor*, between 1 and 3 (usually simplified to 1);
- $V_T$  is the *voltage equivalent of temperature*, around 26 millivolts.

Of course, the breakdown effect experienced by the diode for  $v_D(t) \ll 0$  isn't taken into account in Schottky's law. We usually take approximated models where the diode starts acting like a voltage source if the reverse current conduction constraint is satisfied.



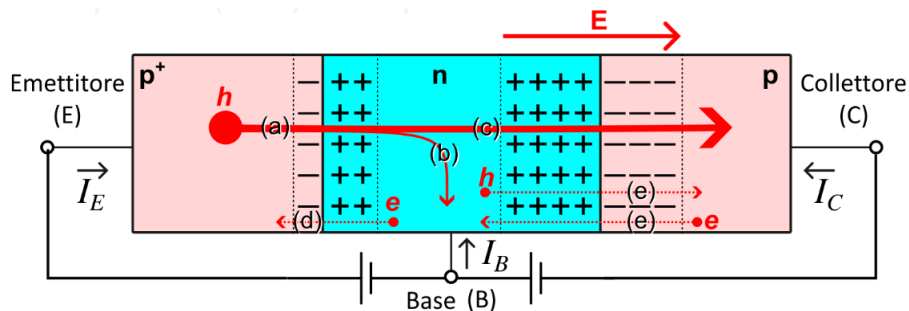
Diodes can be *modeled* in a variety of simplified ways:

- **Ideal switch:** when voltage drop is negative, current is 0; when current is positive, the voltage drop is 0;
- **Ideal switch with offset:** when voltage drop is less than  $V_\gamma$ , the current is 0; when current is positive, the voltage drop is  $V_\gamma$ ;

- **Ideal switch** with *dynamic resistance*: when voltage drop is less than  $V_\gamma$ , the current is 0; when voltage drop is more than  $V_\gamma$ , the current follows a resistive law. This is equivalent to applying a small series resistance when the ideal diode is conducting.

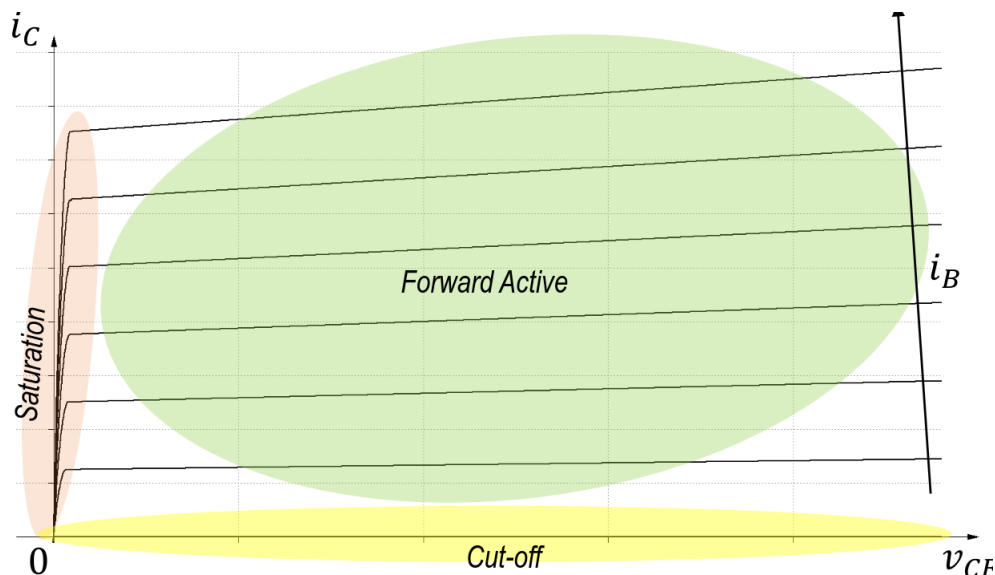
The way circuits are solved by using approximate models is by first making an *assumption* about the state of a diode (conducting or not), and then checking that assumption. Since the models have 2 states, for a diode we can assume that if it's not one state, it's the other. For  $n$  diodes we have to make  $n$  choices, and thus explore a  $2^n$  assumption space.

**BJT transistors** are active semiconductor components, commonly used as amplifiers or current controlled switches. They also are a non-linear component, built from 2 PN junctions put head-to-head (which we know on their own made diodes). Two types of BJT transistors exist: PNP and NPN. NPNs are more common.



The BJT transistor is *tripole* made up of:

- **Emitter**: emits charge carriers scaled by the base current;
- **Base**, used to control the device via current;
- **Collector**, collects charge carriers emitted by the emitter.



BJTs work in 3 main regions:

- **Forward active** region, where the collector current is  $\beta i_B$ , so a direct scaling of the base current by a gain factor  $\beta$ . The voltage drop of the base on the emitter is taken with the usual diode law as  $V_\gamma \approx 0.7V$ . A *reverse active* region also exists but is usually undesirable;

- **Cut-off** region, where the collector current is 0 and the BJT acts as an open switch;
- **Saturation** region, where the  $v_{CE}$  between collector and emitter is approximately 0 (around  $0.1 \sim 0.2V$ ). In this region the collector cannot give more current and the BJT acts as a closed switch.

**MOS transistors** are also active semiconductor components, like BJTs, commonly used as amplifiers or current controlled switches. They use a different technology to obtain basically the same characteristics: historically they were proposed around the time BJTs started becoming popular, but weren't implemented until much later.

The main difference is that MOS transistors act as *voltage controlled*, instead of *current controlled* devices.

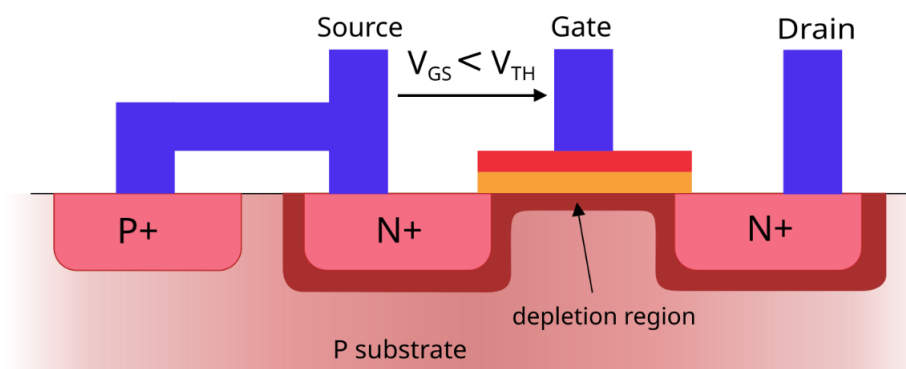
MOSes consist, briefly, of three superimposed layers:

- A base layer of doped semiconductor, which is usually silicon (Si) doped P;
- An intermediate oxide layer, which historically was silicon oxide ( $SiO_2$ ) and today is mainly polysilicate. In general, we assume that the oxide is a perfect insulator (in reality, it never is, but this approximation is useful to us);
- An upper metal layer, which historically was aluminum (Al).

There are then two additional electrodes, connected to N-doped regions in the P-type substrate. We call these electrodes **source** and **drain**. At this point, the bottom electrode  $G$  is called **gate**.

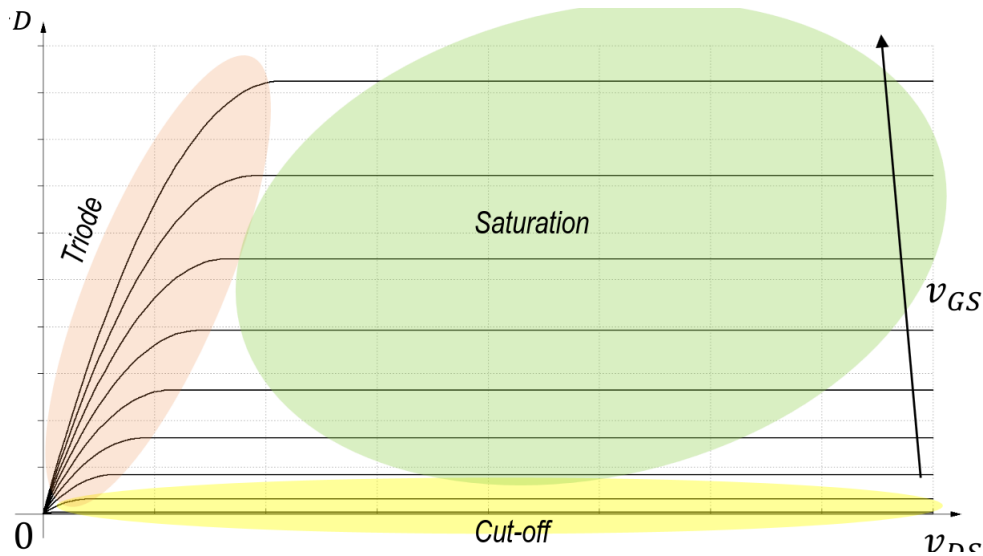
We can see that between the P-type substrate and the two N-type electrodes (source and drain), *diodes* are essentially formed, preventing current from flowing from the source and drain into the substrate.

For the MOSFET to operate, therefore, the N-doped regions need to extend beneath (or at least slightly overlap) the oxide. We call the distance  $L$  between the two N-doped regions the *gate length*.



So, we can say that a MOS transistor is another *tripole* made up of:

- **Source**: emits charge carriers scaled by the base voltage;
- **Gate**, used to control the device via voltage, it is always isolated from drain and source;
- **Drain**, collects charge carriers emitted by the emitter.



MOSes work in 3 main regions:

- **Cut-off** region: when the  $v_{GS}$  voltage between gate and source is lower than a certain  $v_{th}$  threshold voltage, the drain current is zero and the MOS behaves as an open circuit;
- **Triode** or *linear* region: when the  $v_{GS}$  voltage is higher than  $v_{th}$ , but  $v_{DS}$  is less than  $v_{GS} - v_{th}$ , that is:

$$v_{GS} > v_{th}, \quad v_{DS} < v_{GS} - v_{th}$$

then we are in a linear region where current scales mostly directly with voltage applied to  $v_{DS}$ . The exact law is:

$$i_D = \mu C_{ox} \frac{W}{L} \left( (v_{GS} - v_{th})v_{DS} - \frac{v_{DS}^2}{2} \right)$$

where  $C_{ox}$  is the oxidation coefficient of the oxide (usually a polysilicate), and  $W$  and  $L$  are respectively the width and the length of the MOS gate channel;

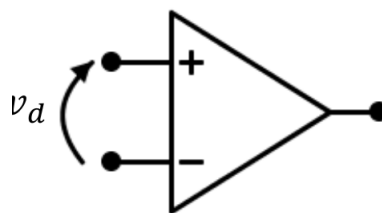
- **Saturation** region: when the  $v_{GS}$  voltage is higher than  $v_{th}$ , but  $v_{DS}$  is higher than  $v_{GS} - v_{th}$ , that is:

$$v_{GS} > v_{th}, \quad v_{DS} > v_{GS} - v_{th}$$

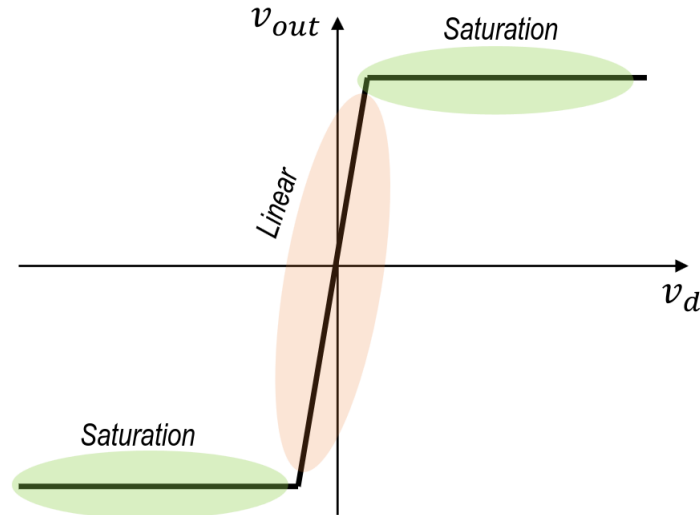
then we are in saturation, where the current scales directly with  $v_{GS}$  independently of  $v_{DS}$ . The exact law is:

$$i_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})^2$$

**Operational amplifiers** are complex components comprised of DC-coupled differential amplifiers with very high gain values.



What OP amps basically do is take the difference in voltage between two terminals (usually labeled + and -) and amplify that difference with an extremely high open loop gain ( $G_{OL}$ ).



This means the output characteristic is:

$$v_{out} = G_{OL}(v^+ - v^-) = G_{OL}v_D$$

Let us note that, even if  $G_{OL}$  is in the order of several millions, an OP amp is a powered device which requires power and ground rails, and that cannot return voltages lower than the ground rail or higher than the power rail (a phenomenon known as *saturation*).

They are most commonly used as a **linear amplifier**:

- With a negative feedback network;
- Forced to work in linear region;
- Lower final gain, larger bandwidth, stability.

Or used as a **comparator**:

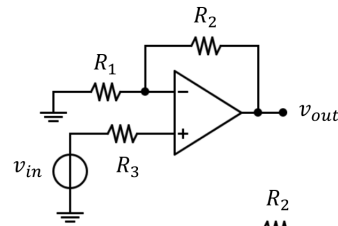
- With a positive feedback network (or no feedback);
- Forced to work in saturation regions.

They are widely used in many embedded systems.

Below, a few common circuits with OP amps are noted.

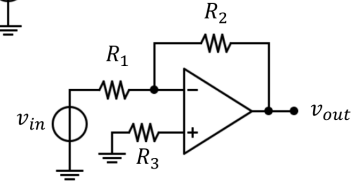
- Non inverting amplifier

- $v_{out} = v_{in} \left(1 + \frac{R_2}{R_1}\right)$ ,  $R_{in} = \infty$ ,  $R_{out} = 0$
- $R_3$  used to reduce unwanted output offsets



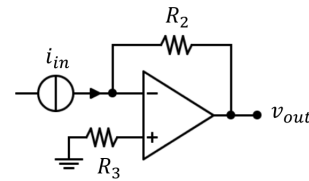
- Inverting amplifier

- $v_{out} = v_{in} \left(-\frac{R_2}{R_1}\right)$ ,  $R_{in} = R_1$ ,  $R_{out} = 0$
- Note the non ideal (low) input resistance



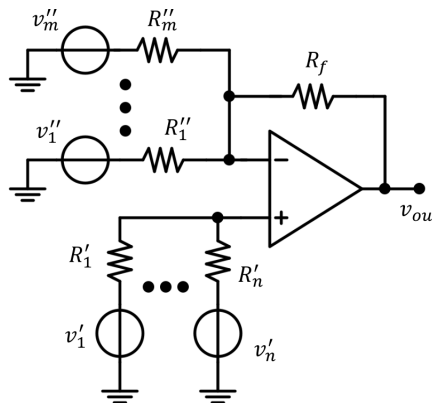
- Transresistance amplifier

- $v_{out} = -R_2 i_{in}$ ,  $R_{in} = 0$ ,  $R_{out} = 0$



We note that transresistance amplifiers will be useful to us when we need to implement analog converters.

**Generalized analog adders / subtractors** can be designed via OP amps by connecting inputs we want to add to the non inverting input and inputs we want to subtract to the inverting input.



Let's make sure to clearly define what we're after before delving into the implementation details. We ideally want a network based on a negative feedback OP amp amplifier, with a single  $R_f$  resistance as the feedback network.

- On the non-inverting input, we'll connect in parallel  $n$  voltage generators  $v'_1, v'_2, \dots, v'_n$ , with series resistances of  $R'_1, R'_2, \dots, R'_n$ ;
- On the inverting input, we'll connect in parallel  $m$  voltage generators  $v''_1, v''_2, \dots, v''_m$ , with series resistances of  $R''_1, R''_2, \dots, R''_m$ .

We want the output of this network to perform the following operation on the voltage sources, taken as inputs:

$$V_{out} = \sum_{i=1}^n A'_i v'_i - \sum_{i=1}^m A''_i v''_i$$

- Let's recall that the inverting input gain can be taken as:

$$V_{out_j}'' = -\frac{R_f}{R_j''} v_j''$$

- Whereas the non-inverting input gain can be taken as:

$$V_{out_j}' = \left(1 + \frac{R_f}{\prod_{i=1}^m R_i''}\right) V_j'$$

The idea is to choose the resistances  $R_i'$  and  $R_i''$  such that the expression of  $V_{out}$  above holds.

- Choosing the resistances on the inverting side is really easy as we can simply say:

$$R_i'' = \frac{R_f}{A_i'}$$

and in the inverting input gain equation we get the correct gain;

- For the inverting side, we first compute the voltage on node  $V^+$  through Millman's theorem (theorem 1.2, of which we gave a proof in section 1.2.5):

$$V^+ = \frac{\sum_{i=1}^n \frac{v_i'}{R_i'}}{\sum_{i=1}^n \frac{1}{R_i'}}$$

From this equation we see that a good starting point is setting:

$$R_i' = \frac{R_f}{A_i''}$$

so that the gains at the numerator are correct:

$$V^+ = \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'}$$

Now we use the non-inverting input gain to compute the output of the non-inverting side:

$$\begin{aligned} V_{out}' &= V^+ \left(1 + \frac{R_f}{\prod_{i=1}^m R_i''}\right) = \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + \frac{R_f}{\prod_{i=1}^m \frac{1}{R_i''}}\right) \\ &= \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + R_f \sum_{i=1}^m \frac{1}{R_i''}\right) = \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + \sum_{i=1}^m A_i''\right) \end{aligned}$$

At this point we impose the output be equal to the positive side of the original output equation:

$$\frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + \sum_{i=1}^m A_i''\right) = \sum_{i=1}^n A_i' v_i' \implies \frac{1 + \sum_{i=1}^m A_i''}{\sum_{i=1}^n A_i'} = 1$$

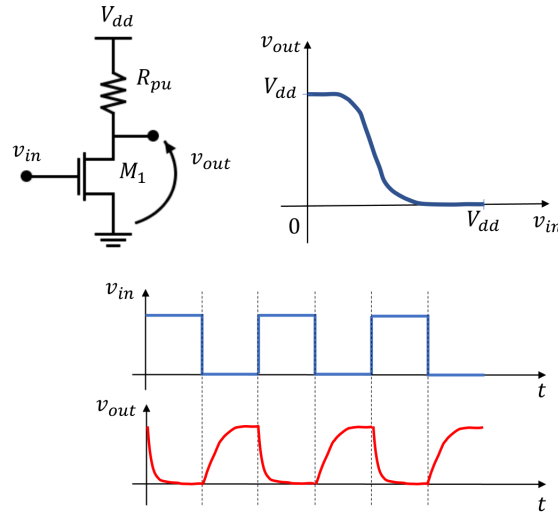
This last equation we obtained is what we have to satisfy in order to get the correct resistor values. It's pretty simple to make the result right by just adding a resistance on the inverting or non-inverting path based on the value of the left hand side, which we'll call  $\alpha$ :

$$\alpha = \frac{1 + \sum_{i=1}^m A_i''}{\sum_{i=1}^n A_i'}, \quad \begin{cases} A_{n+1}' = 1 - \sum_{i=1}^n A_i' + \sum_{i=1}^m A_i'' \\ A_{m+1}'' = -1 + \sum_{i=1}^n A_i' - \sum_{i=1}^m A_i'' \end{cases}$$

## 2.1 Logic gates

We can move on to digital electronics concept, by introducing **logic gates**. A logic gate is an electronic component which performs some sort of operation on digital signals.

A simple example of a logic gate is an **inverter**. The simplest inverter is given by a *resistive load* inverter.



What happens in the circuit is that if we raise  $v_{in}$ , and thus  $v_{GS}$ , above the threshold voltage  $v_{th}$ , then we get conduction between the drain and source of the MOS transistors, and the circuit acts as an open switch. If we then sample  $v_{out}$ , we see the voltage after the fall on  $R_{pu}$  (which will be close to zero).

In the other two graphs we see two non idealities of this circuit:

- **Asymmetric** response: the falling edge is much sharper than rising edge, as the  $R_{DS}$  of the NMOS is much smaller than  $R_{pu}$ ;
- We have **static** power consumption in the on state (when output is at logic 0), as current flows from drain to source:

$$P = \frac{v_{DD}^2}{R_{DS} + R_{pu}}$$

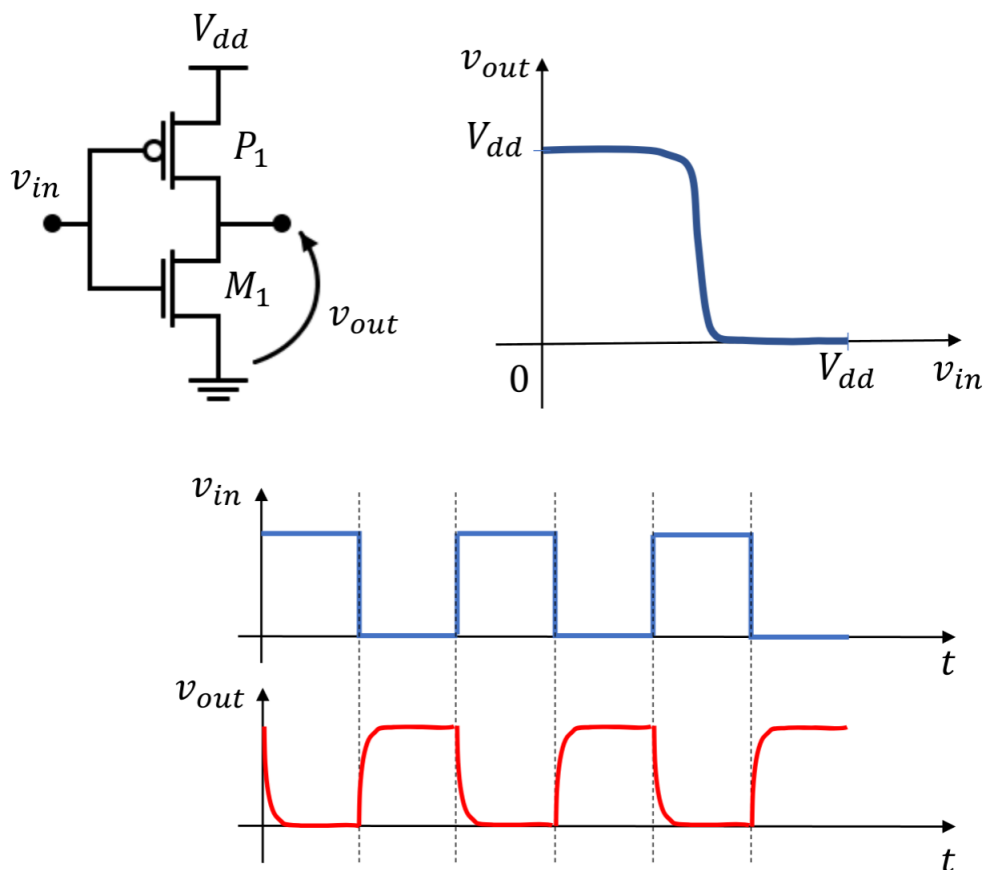
- We also have a certain amount of **dynamic** power consumption, given by the transients happening at state changes. These rely on equivalent capacitances of wires and MOS transistors, and will be studied in more detail later.

We can then see that we need a better design, at least in term of response alignment, and absolutely without static power consumption.

## 3 Lesson 25-09-26

In the last lesson we introduced the resistive load inverter. We said that the main problem of such a design is the asymmetric response, and static power consumption (as in the output off state we have a closed path from power source to ground, draining power on a resistor). Today we try to solve those problems.

**CMOS inverter:** a better inverter can be designed by using **Complementary MOS (CMOS)** designs.



We see that in such a design:

- The response is **symmetric** (or mostly symmetric, we can act on the gate sizes to account for mobility effects);
- There is no **static** power consumption (in the ideal case, some leakage does happen);
- **Dynamic** power is present and measurable based on equivalent capacitance of the wire and the MOS transistors:

$$P \approx C_{e1} f V^2$$

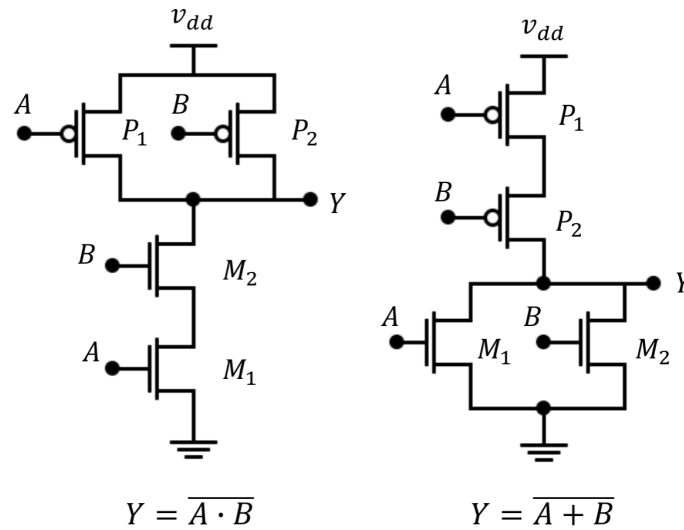
This design is very common in digital circuits nowadays, as it performs much better than the resistive load inverter.

**CMOS logic:** other logic functions are needed to implement a logic family. For CMOS technology, we can use combinations of transistors in series or in parallel to implement any kind of logic.

In particular:

- In the pull-down branch:
  - Series NMOS implements a NAND
  - Parallel NMOS implements a NOR
- In the pull-up branch:
  - Series PMOS implements a NOR
  - Parallel PMOS implements a NAND

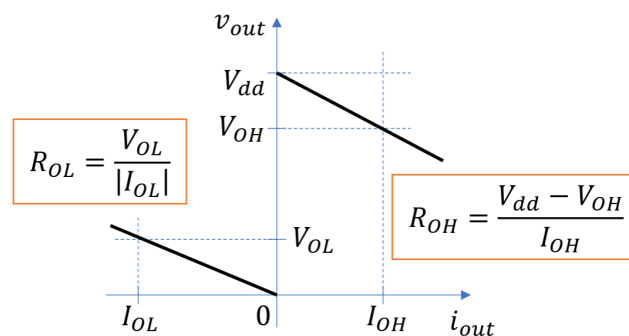
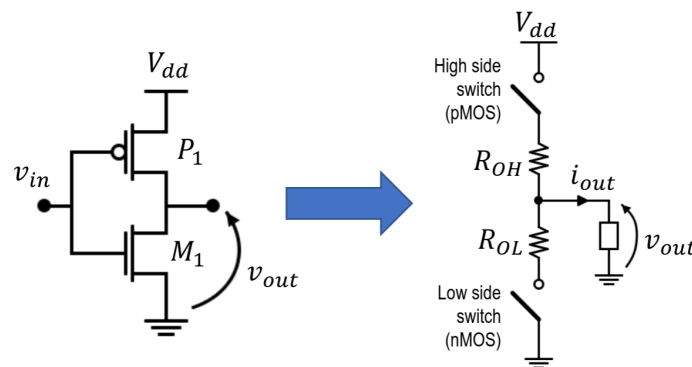
We show a figure of the synthesis of a NAND and NOR circuit in CMOS technology:



**Logic gate parameters:** we can define input and output parameters for logic gates, regarding both *current* and *voltage*.

With regards to **current**, we can model a CMOS inverter converting the MOS transistors to voltage controlled switches (ideal) with series resistances modeling leakage. This way we see we can define the parameters:

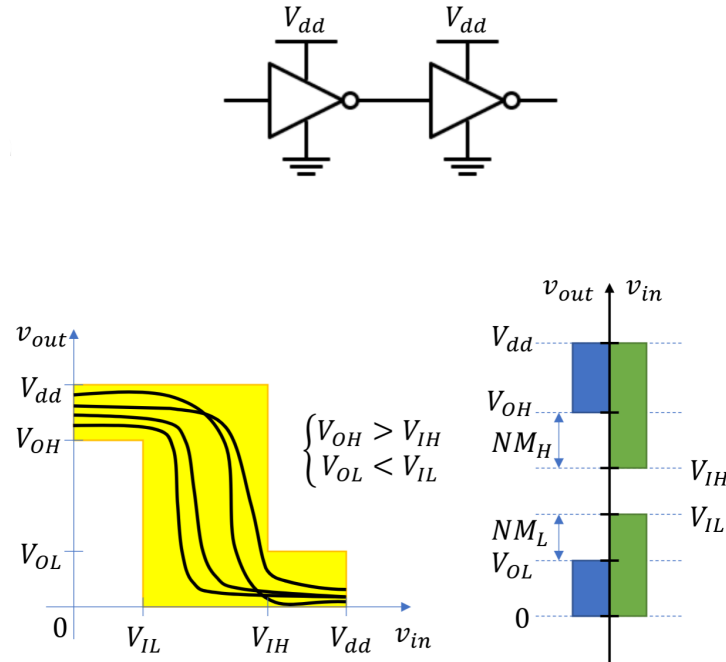
- $V_{OH}$ , which is the lowest output voltage to get a logic high output;
- $V_{OL}$ , which is the highest output voltage to get a logic low output;
- $I_{OH}$ , which is the maximum output current such that the output voltage at logic high is greater than  $V_{OH}$ ;
- $I_{OL}$ , which is the maximum output current such that the output voltage at logic low is less than  $V_{OL}$ .



From these parameters (given by constructors) we can compute the value of the modeling resistances:

$$R_{OL} = \frac{V_{OL}}{|I_{OL}|}, \quad R_{OH} = \frac{V_{dd} - V_{OH}}{I_{OH}}$$

Now, with regards to input and output **voltages** in stacked gate setups, we can consider the following:



Here we have a pair of connected inverters. We are interested on whether output voltages of the first inverter will be correctly interpreted at the input of the second inverter, so we define the following parameters:

- $V_{IH}$ , which is the lowest input voltage recognized as logic high;
- $V_{IL}$ , which is the highest input voltage recognized as logic low;

From this we can derive stability *conditions* for the whole logic family:

$$V_{OH} > V_{IH}, \quad V_{OL} < V_{IL}$$

where we define the following margins:

$$V_{OH} - V_{IH} = NM_H \quad (\text{noise margin high})$$

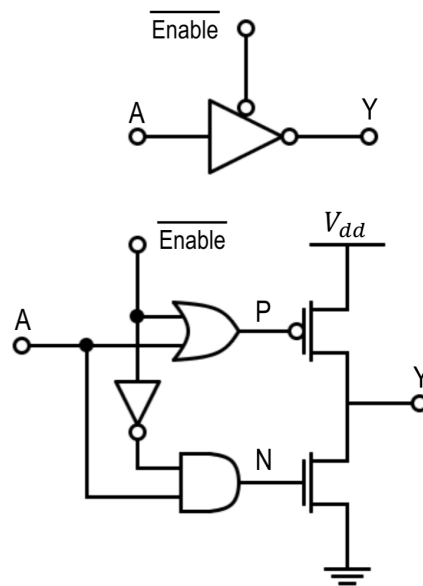
$$V_{IL} - V_{OL} = NM_L \quad (\text{noise margin low})$$

**Tri-state** outputs are needed to avoid logic gates shorting when connected to the same outputs.

In these designs we have an additional Enable input, which can be *active high* or *active low*:

- When Enable is active, the output is driven normally;
- When Enable is NOT active, both the pull-up (high side) and pull-down (low side) transistors are switched OFF, and the output line is left floating. At this point the voltage level (and consequently the logic level) is set by some other circuit, or remains undetermined.

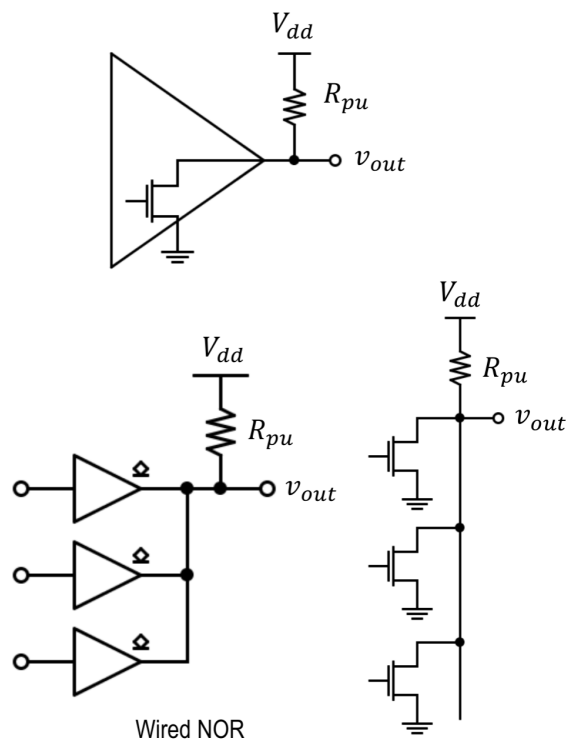
Below, we show the symbol of a tri-state inverter and a possible CMOS implementation.



As we said, tri-state outputs are useful when multiple logic gates drive the same line, for instance in a shared bus. All drivers must be in high impedance except one: it is vital to guarantee the above condition to avoid short circuits.

**Open collector** is another technology which allows to drive the same data line, even without knowing the voltage supply value.

In this setup, there is no pull-up transistor at the output: we can only set the output to logic 0. To set a logic 1, we switch off all pull-down transistors, and use an external pull-up resistor.



This is useful to drive the same line with multiple logic gates:

- No possible short circuit: when all logic gates set a logic 1 the output is at logic 1;
- When at least one logic gate sets a logic 0, the output goes at logic 0. This makes open collector a NOR logic function.

This is slightly easier to implement than the earlier tri-state solution, as we don't have to deal with high impedance or short circuit issues.

Of course, we can imagine an *open drain* setup which works the same, only leaving the pull-down transistor outside of the design and pulling *up* through an external resistor.

### 3.1 Combinatorial circuits

**Combinatorial circuits** are the simplest we can see with digital logic circuits. In a combinatorial logic circuit, the logic output depends on the current value of the inputs only (no state).

A combinatorial circuit can be described through a **truth table**, which is a list all possible input combinations, and corresponding output values.

Many ways exist to synthesize a combinatorial circuit:

- The main idea is generally to generate the circuit as a **SoP** (*Sum of Products*), minimizing the number of products, and minimizing the number of literals in each product (which corresponds to minimizing components).
- A possible tool is given by *Karnaugh maps*;
- Another possibility is the *Quine-McCluskey algorithm* for more inputs;
- Often, we just stick to logic synthesis *software tools*, which implement these and other advanced algorithms.

**Multiplexers** are circuits which allow to select an input and output just one of them, between different inputs. A one bit multiplexer with 2 inputs has this boolean law:

$$y = a \cdot s + b \cdot \bar{s}$$

where  $a$  and  $b$  are the inputs, and  $s$  is the selection signal.

For multiplexers on multiple bits, we repeat the same structure multiple times; For multiplexers with more than two inputs, we need more than one selection signal (usually built hierarchically with log number of inputs smaller multiplexers). Multiplexers with more than two inputs on multiple bits are built as a combination of the above two cases.

**Decoders** decode the binary representation of a number, on  $n$  bits, to the same number in one-hot representation, on  $2^n$  bits.

**Encoders** do the reverse job of converting a one-hot representation on  $2^n$  bits in a single  $n$  bit encoded value.

More complex components like *priority encoders*, ... can be built.

**Digital adders / subtractors** are built in combinatorial logic too.

We start from *half adders*: these operate on 1-bit operands, generating a carrying output bit. For *multibit* adders we combine multiple half adders.

Different area and performance can be achieved based on carry computation. For instance, we remember:

- Ripple Carry
- Carry Look Ahead
- Carry Select

*Subtractors* are just implemented using an adder (using 2-complement notation).

### 3.2 Sequential circuits

**Sequential circuits** need a *memory element* to exist somewhere in a circuit. The most popular memory element is given by a pair of active elements in a loop.

- A pair of inverters in a loop provides a memory element, that is an element that is *bistable*: either of the inverters can start out with an high outputs, and the state is converted. The problem of this design is that we can't control it:
- A pair of NANDS in a loop can be used in the same fashion. The input bits that are left out can be used as SET and RESET parameters for the bistable circuit. We call this a **SR latch**.

